

Question-4.3.6

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Question

Passing through the points $(3, 4, -7)$ and $(1, -1, 6)$.

Solution

let Passing through the points $\mathbf{A} = \begin{pmatrix} 3 \\ 4 \\ -7 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 \\ -1 \\ 6 \end{pmatrix}$.

$$\begin{pmatrix} 3 & 4 & -7 \\ 1 & -1 & 6 \end{pmatrix} \mathbf{n} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Writing the augmented matrix:

$$\begin{pmatrix} 3 & 4 & -7 & 1 \\ 1 & -1 & 6 & 1 \end{pmatrix} R_1 \leftarrow R_1 - 3R_2 \begin{pmatrix} 0 & 7 & -25 & -2 \\ 1 & -1 & 6 & 1 \end{pmatrix} \quad (1)$$

$$R_1 \leftrightarrow R_2 \begin{pmatrix} 1 & -1 & 6 & 1 \\ 0 & 7 & -25 & -2 \end{pmatrix} \quad (2)$$

$$n_1 - n_2 + 6n_3 = 1 \quad (3)$$

$$7n_2 - 25n_3 = -2 \quad (4)$$

From third equation:

$$n_2 = \frac{-2 + 25n_3}{7} \quad (5)$$

Solution

Substitute in second equation:

$$n_1 = \frac{5 - 17n_3}{7} \quad (6)$$

Let $n_3 = t$, then

$$\mathbf{n} = \begin{pmatrix} \frac{5-17t}{7} \\ \frac{-2+25t}{7} \\ t \end{pmatrix} \quad (7)$$

let $t = 1$, multiply with 7 to clear denominators:

$$\mathbf{n} = \begin{pmatrix} -12 \\ 23 \\ 7 \end{pmatrix} \quad (8)$$

Thus, the equation of the line is: $(-12 \ 23 \ 7) \mathbf{x} = 7$

Line Through 3D Points A and B

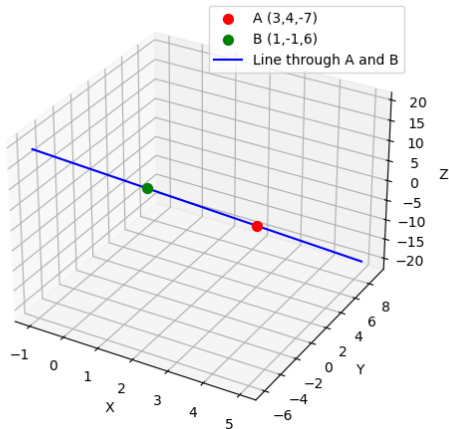


Figure:

```
// code.c
#include <stdio.h>

void get_points_and_line(double* A, double* B, double* n, double*
    k) {
    // Input points:
    A[0] = 3; A[1] = 4; A[2] = -7;
    B[0] = 1; B[1] = -1; B[2] = 6;
    // Precalculated normal vector and constant for these points
    n[0] = -12; n[1] = 23; n[2] = 7;
    *k = 7;
}
```

```
# Code by GVV Sharma  
# Modified for Problem Solution  
# Released under GNU GPL  
# Calculating area enclosed between curves  
# call.py  
import ctypes  
import numpy as np  
  
lib = ctypes.CDLL('./code.so')
```

```
def get_points_and_line():
    A = np.zeros(3, dtype=np.float64)
    B = np.zeros(3, dtype=np.float64)
    n = np.zeros(3, dtype=np.float64)
    k = ctypes.c_double()
    lib.get_points_and_line.argtypes = [
        ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.
            c_double),
        ctypes.POINTER(ctypes.c_double), ctypes.POINTER(ctypes.
            c_double)
    ]
    lib.get_points_and_line(
        A.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
        B.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
        n.ctypes.data_as(ctypes.POINTER(ctypes.c_double)),
        ctypes.byref(k)
    )
```



```
    return A, B, n, k.value

if __name__ == "__main__":
    A, B, n, k = get_points_and_line()
    print("Point A:", A)
    print("Point B:", B)
    print("Normal Vector n:", n)
    print("Constant k:", k)
```

```
import numpy as np
import matplotlib.pyplot as plt
from call import get_points_and_line

# Extract from .so file
A, B, n, k = get_points_and_line()

# Direction vector of the line ( $B - A$ )
d = B - A

# Parameter range for the line
t_vals = np.linspace(-1, 2, 100)
line_points = np.array([A + t * d for t in t_vals])

# Plot
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
```

```
# Plot the two points
ax.scatter(A[0], A[1], A[2], color='red', label='A (3,4,-7)', s
           =50)
ax.scatter(B[0], B[1], B[2], color='green', label='B (1,-1,6)', s
           =50)

# Plot the line passing through A and B
ax.plot(line_points[:,0], line_points[:,1], line_points[:,2],
        color='blue', label='Line through A and B')

ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')
ax.legend()
ax.set_title("Line Through 3D Points A and B")
plt.tight_layout()
plt.show()
```